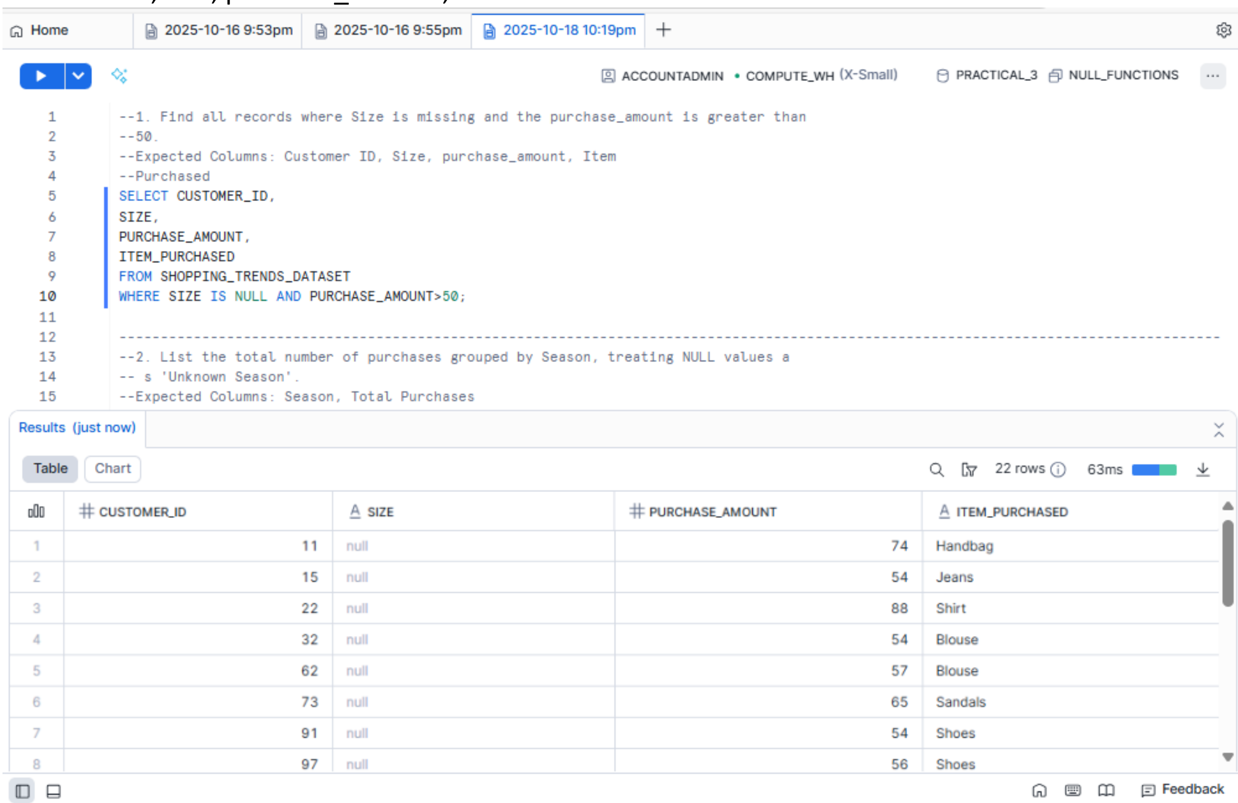
BrightLight Data Analytics Coding Practical
ADVANCE SQL
PRACTICAL-3
Thokozani Sibiya

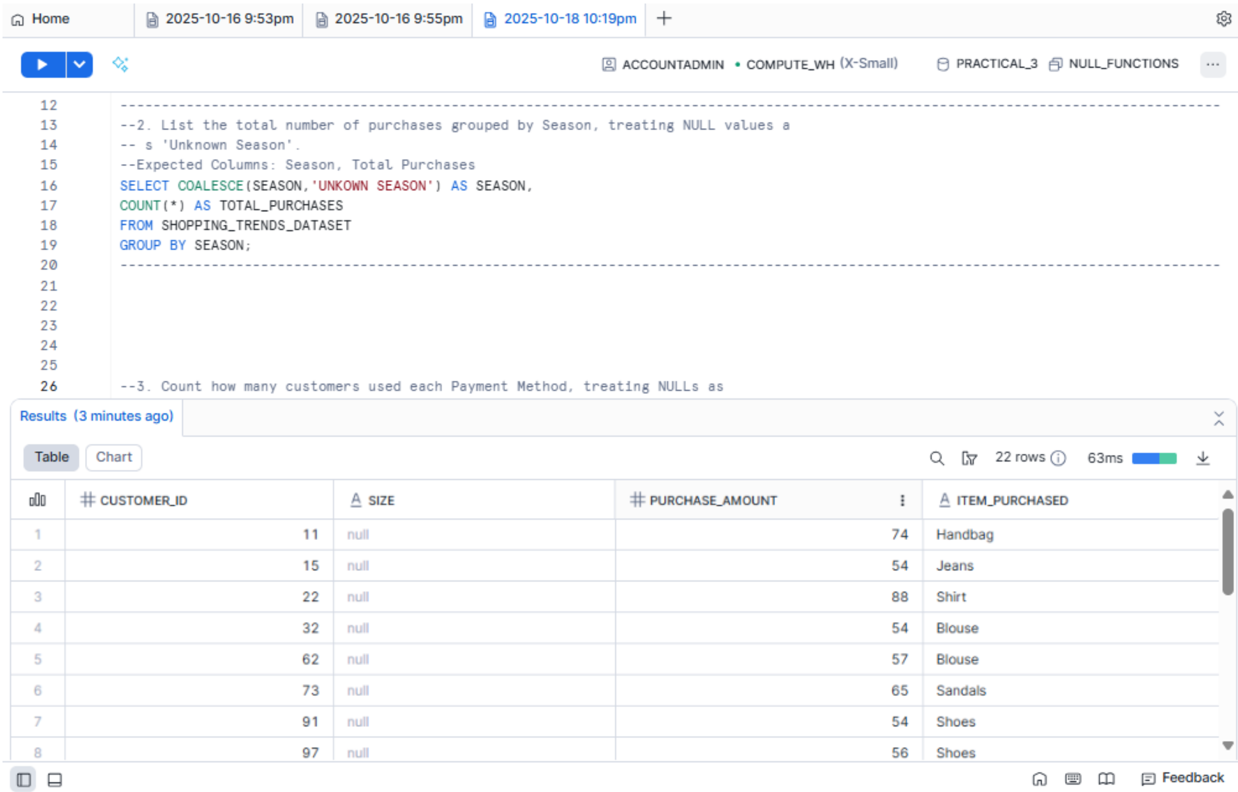
1. Find all records where Size is missing and the purchase_amount is greater than 50. Expected Columns: Customer ID, Size, purchase_amount, Item Purchased



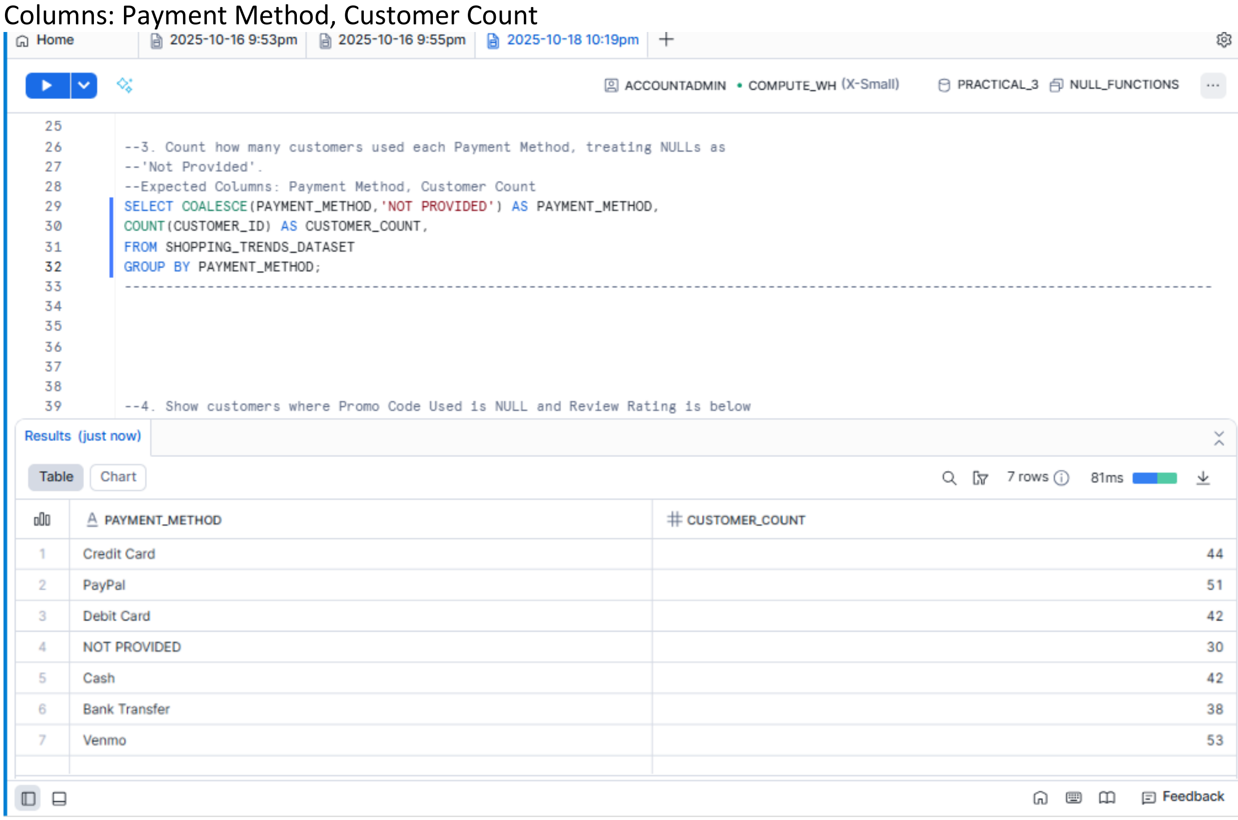

```sql
--1. Find all records where Size is missing and the purchase_amount is greater than
--50.
--Expected Columns: Customer ID, Size, purchase_amount, Item
--Purchased
SELECT CUSTOMER_ID,
SIZE,
PURCHASE_AMOUNT,
ITEM_PURCHASED
FROM SHOPPING_TRENDS_DATASET
WHERE SIZE IS NULL AND PURCHASE_AMOUNT>50;

--------------------------------------------------------------------------------------------------------------
--2. List the total number of purchases grouped by Season, treating NULL values a
-- s 'Unknown Season'.
--Expected Columns: Season, Total Purchases
```

	CUSTOMER_ID	SIZE	PURCHASE_AMOUNT	ITEM_PURCHASED
1	11	null	74	Handbag
2	15	null	54	Jeans
3	22	null	88	Shirt
4	32	null	54	Blouse
5	62	null	57	Blouse
6	73	null	65	Sandals
7	91	null	54	Shoes
8	97	null	56	Shoes

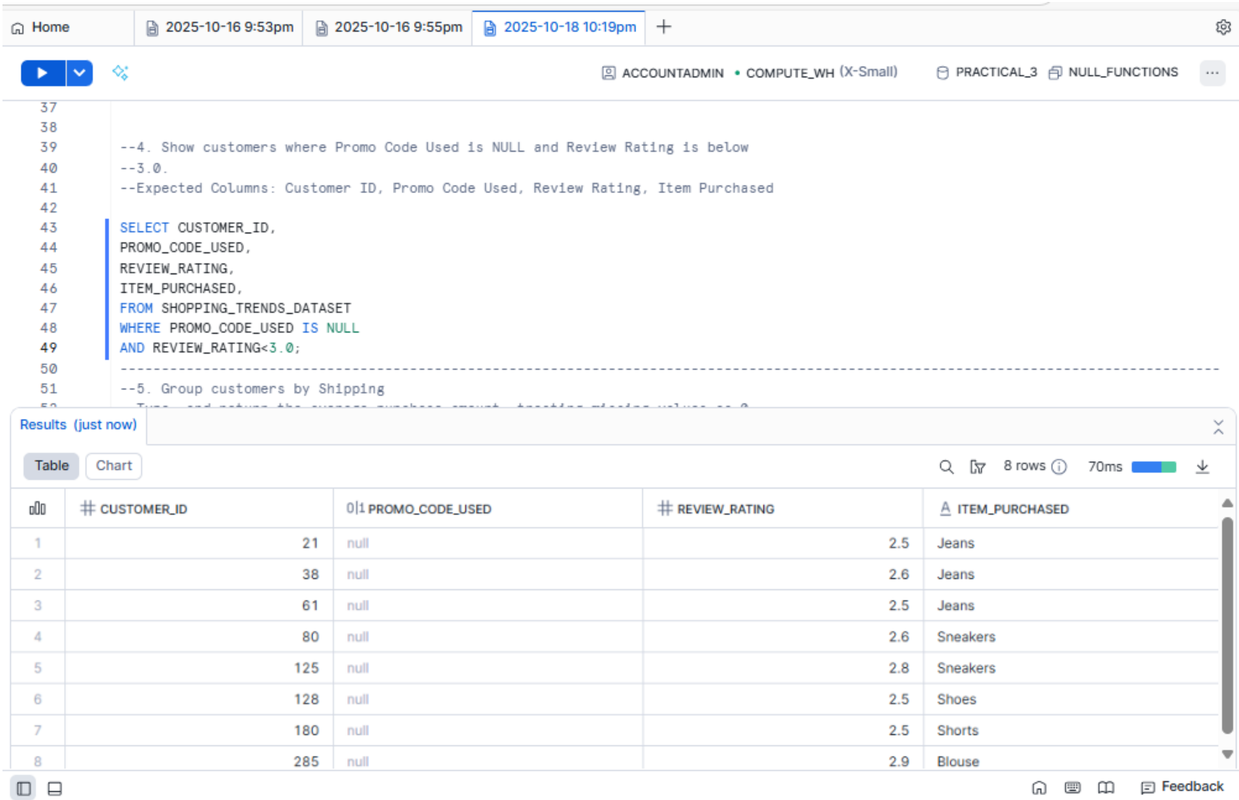
2. List the total number of purchases grouped by Season, treating NULL values a s 'Unknown Season'. Expected Columns: Season, Total Purchases



3. Count how many customers used each Payment Method, treating NULLs as 'Not Provided'. Expected Columns: Payment Method, Customer Count



4. Show customers where Promo Code Used is NULL and Review Rating is below 3.0. Expected Columns: Customer ID, Promo Code Used, Review Rating, Item Purchased



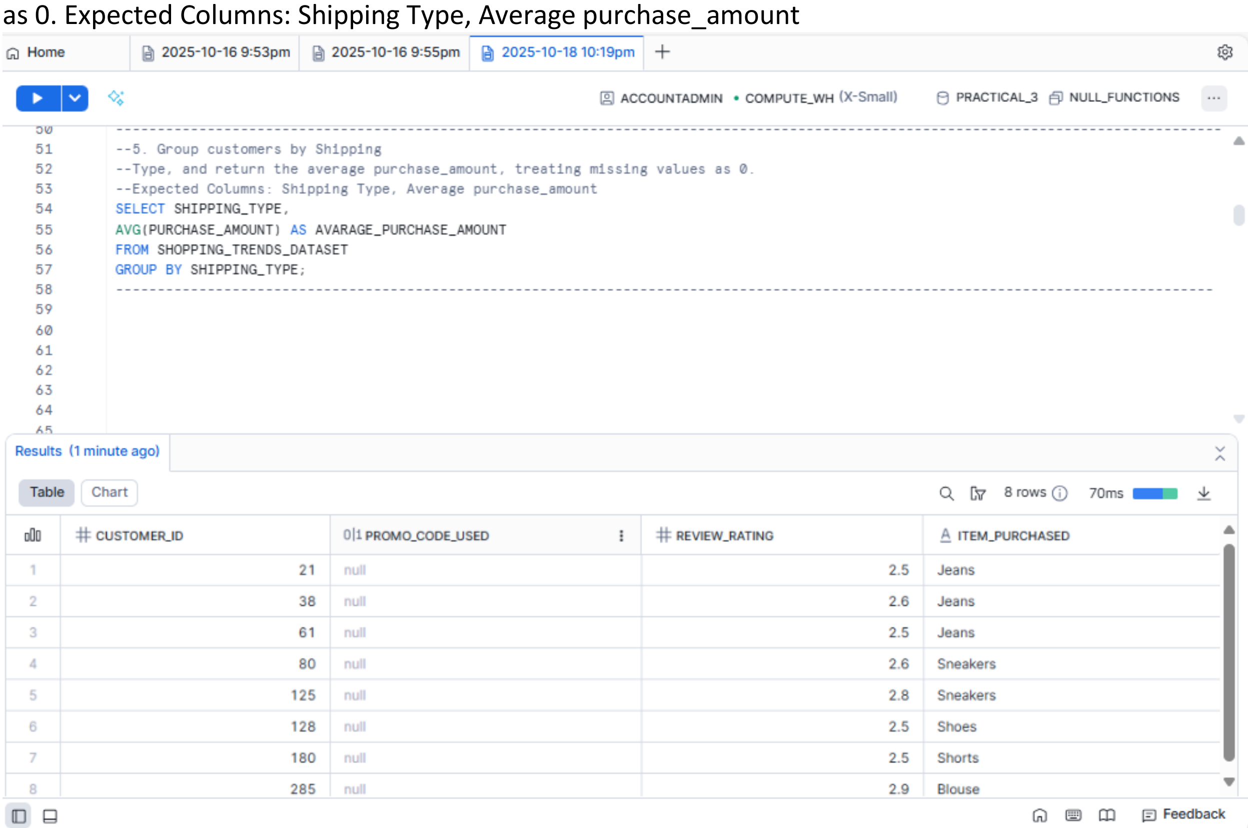

```sql
--4. Show customers where Promo Code Used is NULL and Review Rating is below
--3.0.
--Expected Columns: Customer ID, Promo Code Used, Review Rating, Item Purchased

SELECT CUSTOMER_ID,
PROMO_CODE_USED,
REVIEW_RATING,
ITEM_PURCHASED,
FROM SHOPPING_TRENDS_DATASET
WHERE PROMO_CODE_USED IS NULL
AND REVIEW_RATING<3.0;
---------------------------------------------------------------------------------------------------------------------------------------
--5. Group customers by Shipping
```

Results (just now)

Table Chart 8 rows 70ms

	CUSTOMER_ID	PROMO_CODE_USED	REVIEW_RATING	ITEM_PURCHASED
1	21	null	2.5	Jeans
2	38	null	2.6	Jeans
3	61	null	2.5	Jeans
4	80	null	2.6	Sneakers
5	125	null	2.8	Sneakers
6	128	null	2.5	Shoes
7	180	null	2.5	Shorts
8	285	null	2.9	Blouse

5. Group customers by Shipping Type, and return the average purchase_amount, treating missing values as 0. Expected Columns: Shipping Type, Average purchase_amount



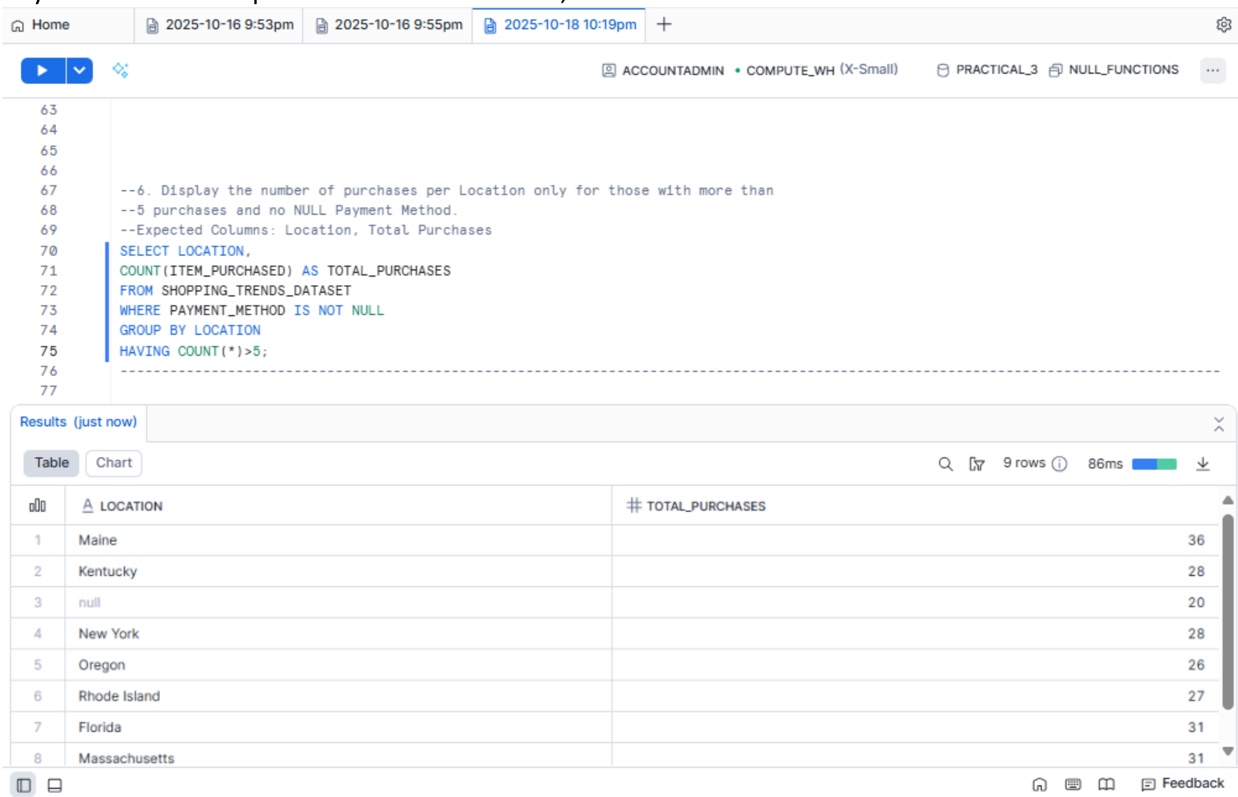

```sql
--5. Group customers by Shipping
--Type, and return the average purchase_amount, treating missing values as 0.
--Expected Columns: Shipping Type, Average purchase_amount
SELECT SHIPPING_TYPE,
AVG(PURCHASE_AMOUNT) AS AVARAGE_PURCHASE_AMOUNT
FROM SHOPPING_TRENDS_DATASET
GROUP BY SHIPPING_TYPE;
---------------------------------------------------------------------------------------------------------------------------------------
```

Results (1 minute ago)

Table Chart 8 rows 70ms

	CUSTOMER_ID	PROMO_CODE_USED	REVIEW_RATING	ITEM_PURCHASED
1	21	null	2.5	Jeans
2	38	null	2.6	Jeans
3	61	null	2.5	Jeans
4	80	null	2.6	Sneakers
5	125	null	2.8	Sneakers
6	128	null	2.5	Shoes
7	180	null	2.5	Shorts
8	285	null	2.9	Blouse

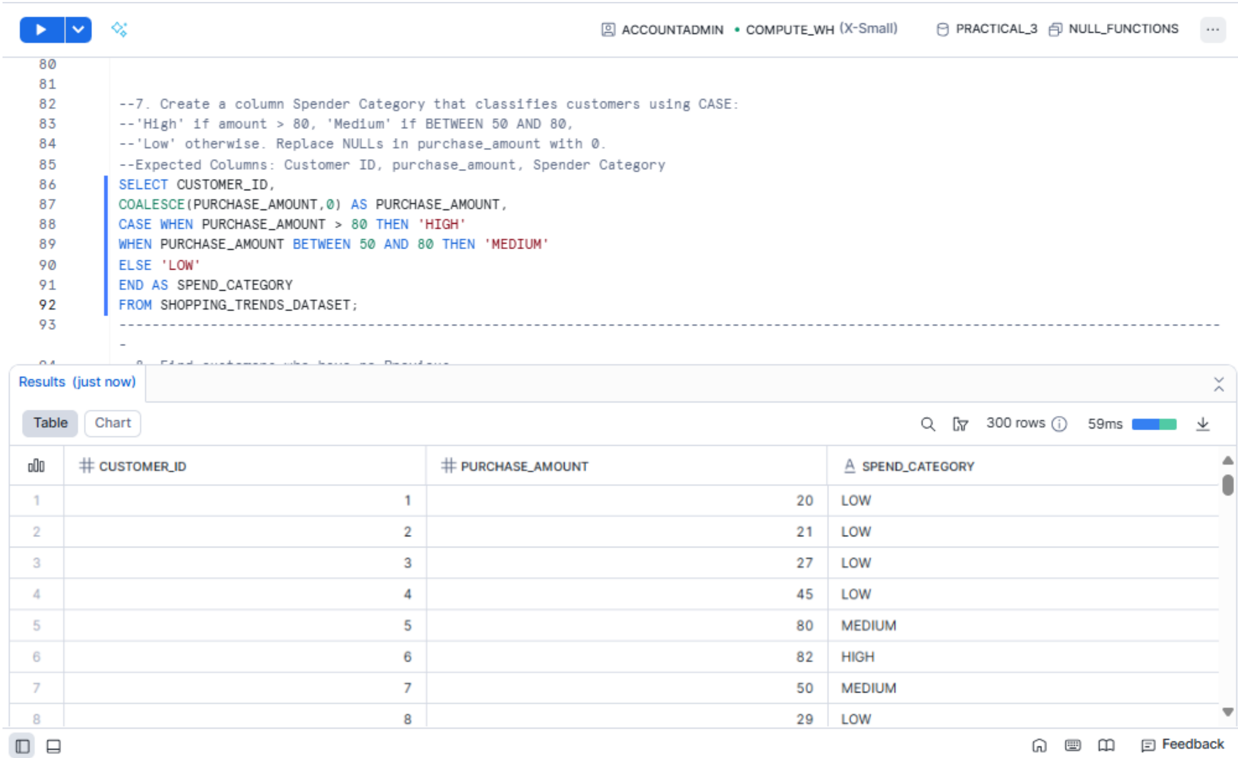
6. Display the number of purchases per Location only for those with more than 5 purchases and no NULL Payment Method. Expected Columns: Location, Total Purchases




```sql
--6. Display the number of purchases per Location only for those with more than
--5 purchases and no NULL Payment Method.
--Expected Columns: Location, Total Purchases
SELECT LOCATION,
COUNT(ITEM_PURCHASED) AS TOTAL_PURCHASES
FROM SHOPPING_TRENDS_DATASET
WHERE PAYMENT_METHOD IS NOT NULL
GROUP BY LOCATION
HAVING COUNT(*)>5;
-----------------------------------------------------------------------------------------------------------------------------------
```


	LOCATION	TOTAL_PURCHASES
1	Maine	36
2	Kentucky	28
3	null	20
4	New York	28
5	Oregon	26
6	Rhode Island	27
7	Florida	31
8	Massachusetts	31

7. Create a column Spender Category that classifies customers using CASE: 'High' if amount > 80, 'Medium' if BETWEEN 50 AND 80, 'Low' otherwise. Replace NULLs in purchase_amount with 0. Expected Columns: Customer ID, purchase_amount, Spender Category



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```sql
--7. Create a column Spender Category that classifies customers using CASE:
--'High' if amount > 80, 'Medium' if BETWEEN 50 AND 80,
--'Low' otherwise. Replace NULLs in purchase_amount with 0.
--Expected Columns: Customer ID, purchase_amount, Spender Category
SELECT CUSTOMER_ID,
COALESCE(PURCHASE_AMOUNT,0) AS PURCHASE_AMOUNT,
CASE WHEN PURCHASE_AMOUNT > 80 THEN 'HIGH'
WHEN PURCHASE_AMOUNT BETWEEN 50 AND 80 THEN 'MEDIUM'
ELSE 'LOW'
END AS SPEND_CATEGORY
FROM SHOPPING_TRENDS_DATASET;
------------------------------------------------------------------------------------------------------
-
```

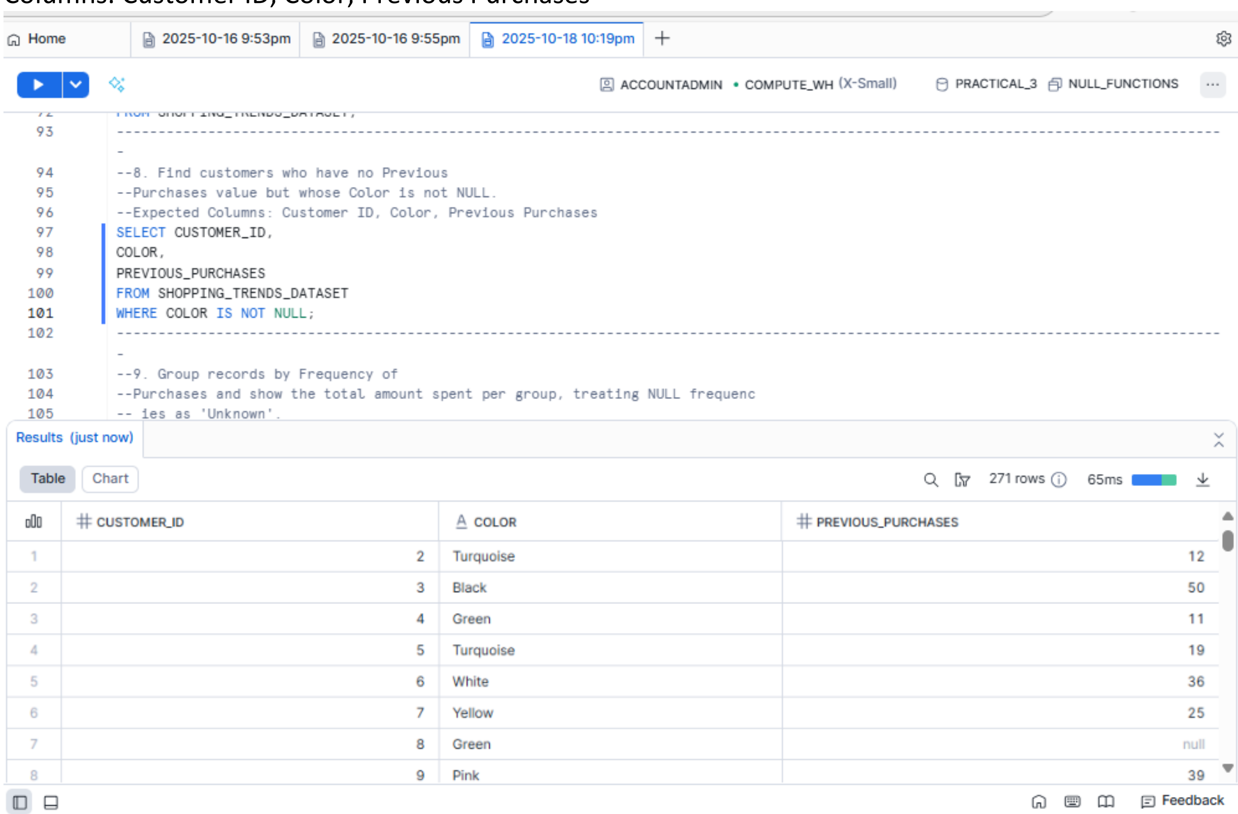
Results (just now)

Table Chart 300 rows 59ms

	CUSTOMER_ID	PURCHASE_AMOUNT	SPEND_CATEGORY
1	1	20	LOW
2	2	21	LOW
3	3	27	LOW
4	4	45	LOW
5	5	80	MEDIUM
6	6	82	HIGH
7	7	50	MEDIUM
8	8	29	LOW

Feedback

8. Find customers who have no Previous Purchases value but whose Color is not NULL. Expected Columns: Customer ID, Color, Previous Purchases



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```sql
------------------------------------------------------------------------------------------------------
-
--8. Find customers who have no Previous
--Purchases value but whose Color is not NULL.
--Expected Columns: Customer ID, Color, Previous Purchases
SELECT CUSTOMER_ID,
COLOR,
PREVIOUS_PURCHASES
FROM SHOPPING_TRENDS_DATASET
WHERE COLOR IS NOT NULL;
------------------------------------------------------------------------------------------------------
-
--9. Group records by Frequency of
--Purchases and show the total amount spent per group, treating NULL frequenc
-- ies as 'Unknown'.
```

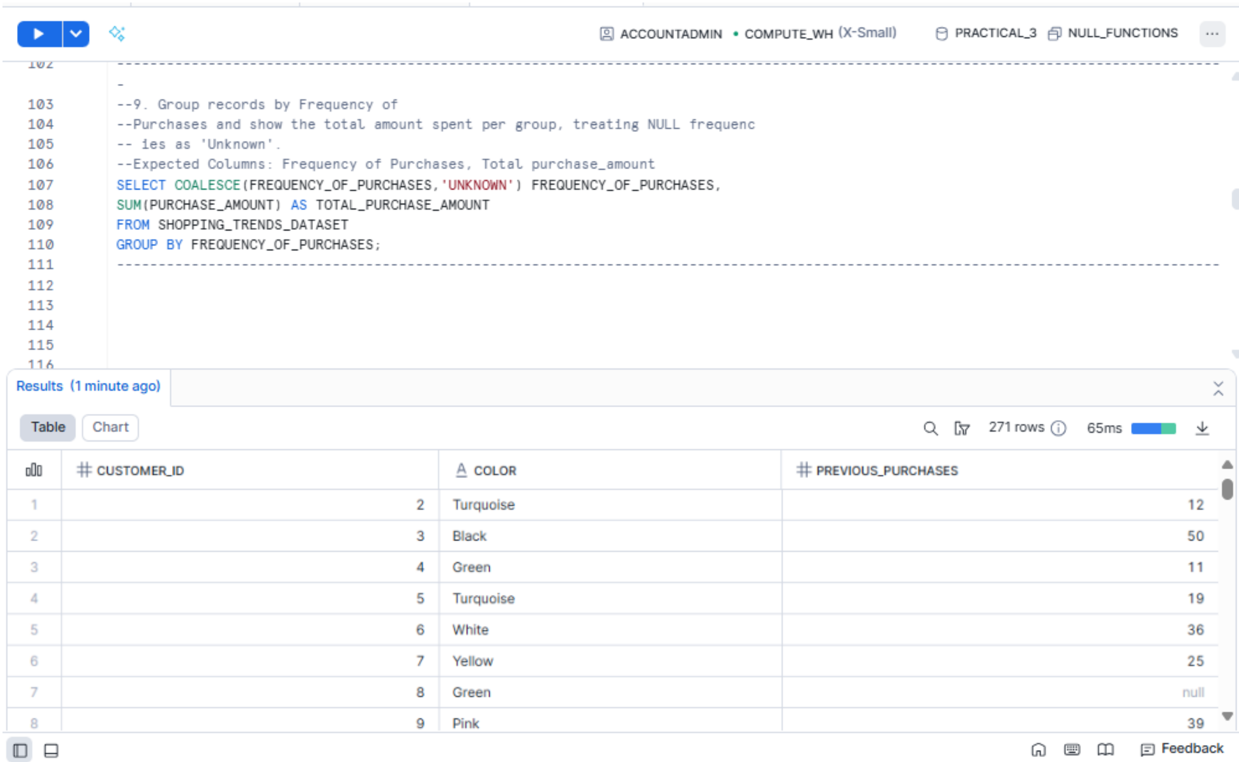
Results (just now)

Table Chart 271 rows 65ms

	CUSTOMER_ID	COLOR	PREVIOUS_PURCHASES
1	2	Turquoise	12
2	3	Black	50
3	4	Green	11
4	5	Turquoise	19
5	6	White	36
6	7	Yellow	25
7	8	Green	null
8	9	Pink	39

Feedback

9. Group records by Frequency of Purchases and show the total amount spent per group, treating NULL frequenc ies as 'Unknown'. Expected Columns: Frequency of Purchases, Total purchase_amount



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```
---------------------------------------------------------------------------
-
--9. Group records by Frequency of
--Purchases and show the total amount spent per group, treating NULL frequenc
-- ies as 'Unknown'.
--Expected Columns: Frequency of Purchases, Total purchase_amount
SELECT COALESCE(FREQUENCY_OF_PURCHASES,'UNKNOWN') FREQUENCY_OF_PURCHASES,
SUM(PURCHASE_AMOUNT) AS TOTAL_PURCHASE_AMOUNT
FROM SHOPPING_TRENDS_DATASET
GROUP BY FREQUENCY_OF_PURCHASES;
---------------------------------------------------------------------------
```

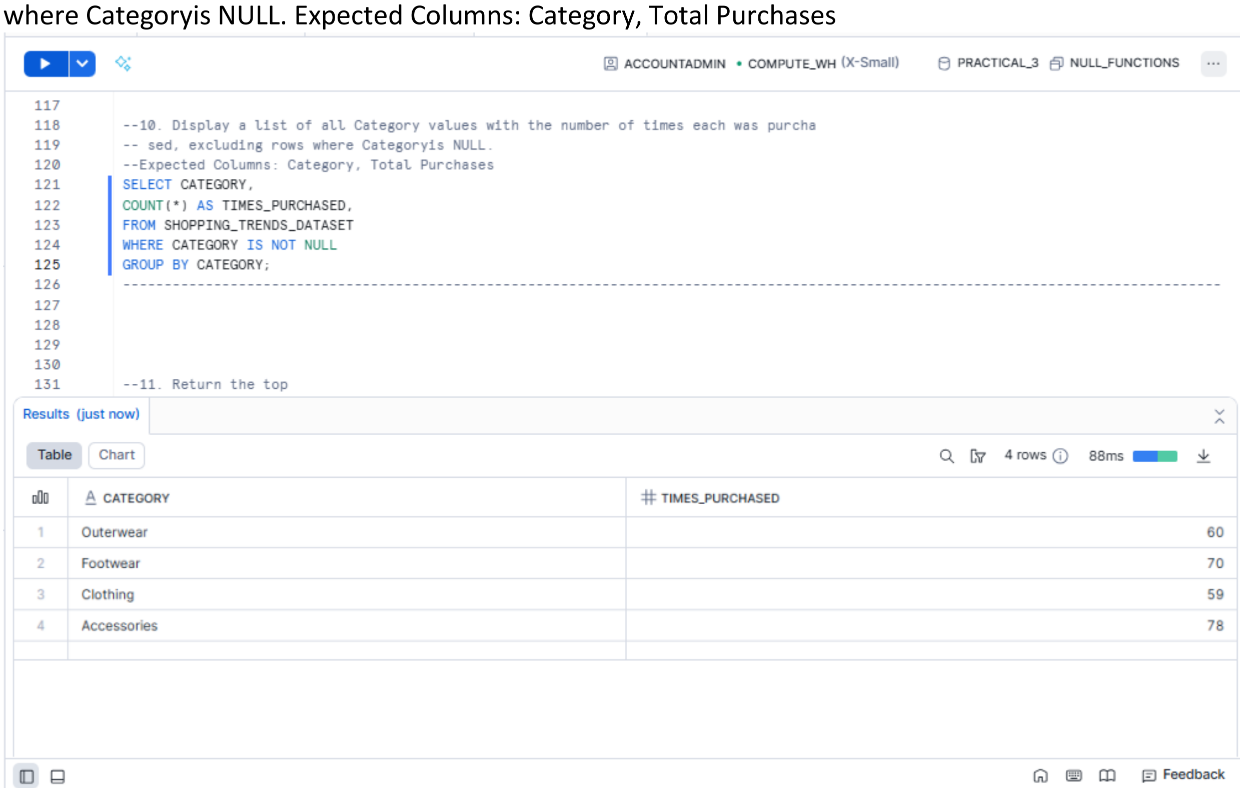
Results (1 minute ago)

Table Chart 271 rows 65ms

	CUSTOMER_ID	COLOR	PREVIOUS_PURCHASES
1	2	Turquoise	12
2	3	Black	50
3	4	Green	11
4	5	Turquoise	19
5	6	White	36
6	7	Yellow	25
7	8	Green	null
8	9	Pink	39

Feedback

10. Display a list of all Category values with the number of times each was purcha sed, excluding rows where Categoryis NULL. Expected Columns: Category, Total Purchases



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```
--10. Display a list of all Category values with the number of times each was purcha
-- sed, excluding rows where Categoryis NULL.
--Expected Columns: Category, Total Purchases
SELECT CATEGORY,
COUNT(*) AS TIMES_PURCHASED,
FROM SHOPPING_TRENDS_DATASET
WHERE CATEGORY IS NOT NULL
GROUP BY CATEGORY;
---------------------------------------------------------------------------




--11. Return the top
```

Results (just now)

Table Chart 4 rows 88ms

	CATEGORY	TIMES_PURCHASED
1	Outerwear	60
2	Footwear	70
3	Clothing	59
4	Accessories	78

Feedback

11. Return the top 5 Locations with the highest total purchase_amount, replacing NULLs in amount with 0. Expected Columns: Location, Total purchase_amount

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```sql
--11. Return the top
--5 Locations with the highest total purchase_amount, replacing NULLs in amount
--with 0.
--Expected Columns: Location, Total purchase_amount
SELECT LOCATION,
SUM (COALESCE(PURCHASE_AMOUNT,'0')) AS TOTAL_PURSHASE_AMOUNT,
FROM SHOPPING_TRENDS_DATASET
GROUP BY LOCATION
ORDER BY TOTAL_PURSHASE_AMOUNT DESC
LIMIT 1;
-------------------------------------------------------------------------------------------------------------------------------------
```

Results (3 minutes ago)

Table Chart 1 row 80ms

	LOCATION	TOTAL_PURSHASE_AMOUNT
1	Maine	2294

Feedback

12. Group customers by Gender and Size, and count how many entries have a NUL L Color. Expected Columns: Gender, Size, Null Color Count

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```sql
--12. Group customers by Gender and Size, and count how many entries have a NUL
-- L Color.
--Expected Columns: Gender, Size, Null Color Count
SELECT GENDER,
SIZE,
COUNT(COLOR) AS NULL_COLOR_COUNT
FROM SHOPPING_TRENDS_DATASET
WHERE COLOR IS NULL
GROUP BY GENDER,SIZE;
-------------------------------------------------------------------------------------------------------------------------------------
--13. Identify all Item Purchased where more than 3 purchases had NULL Shipping
--Type.
--Expected Columns: Item Purchased, NULL Shipping Type Count
```

Results (just now)

Table Chart 5 rows 89ms

	GENDER	SIZE	NULL_COLOR_COUNT
1	Male	null	0
2	Male	M	0
3	Male	L	0
4	Male	S	0
5	Male	XL	0

Feedback

13. Identify all Item Purchased where more than 3 purchases had NULL Shipping Type. Expected Columns: Item Purchased, NULL Shipping Type Count

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```sql
WHERE COLOR IS NULL
GROUP BY GENDER,SIZE;
-----------------------------------------------------------------------------------------------------------------------------
--13. Identify all Item Purchased where more than 3 purchases had NULL Shipping
--Type.
--Expected Columns: Item Purchased, NULL Shipping Type Count
SELECT ITEM_PURCHASED,
COUNT(*) AS NULL_SHIPPING_TYPE
FROM SHOPPING_TRENDS_DATASET
GROUP BY ITEM_PURCHASED
HAVING COUNT(*) >3;
-----------------------------------------------------------------------------------------------------------------------------
```

Results (just now)

Table Chart 11 rows 78ms

	ITEM_PURCHASED	NULL_SHIPPING_TYPE
1	Shirt	26
2	Jeans	25
3	Sweater	27
4	Coat	21
5	Sandals	32
6	Sneakers	39
7	Handbag	20
8	Shoes	26

Feedback

14. Show a count of how many customers per Payment Method have NULL Review Rating. Expected Columns: Payment Method, Missing Review Rating Count

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```sql
--14. Show a count of how many customers per Payment Method have NULL Review
--Rating.
--Expected Columns: Payment Method, Missing Review Rating Count
SELECT PAYMENT_METHOD,
COUNT(REVIEW_RATING) AS MISSING_REVIEW_RATING_COUNT
FROM SHOPPING_TRENDS_DATASET
WHERE REVIEW_RATING IS NULL
GROUP BY PAYMENT_METHOD;
-----------------------------------------------------------------------------------------------------------------------------
```

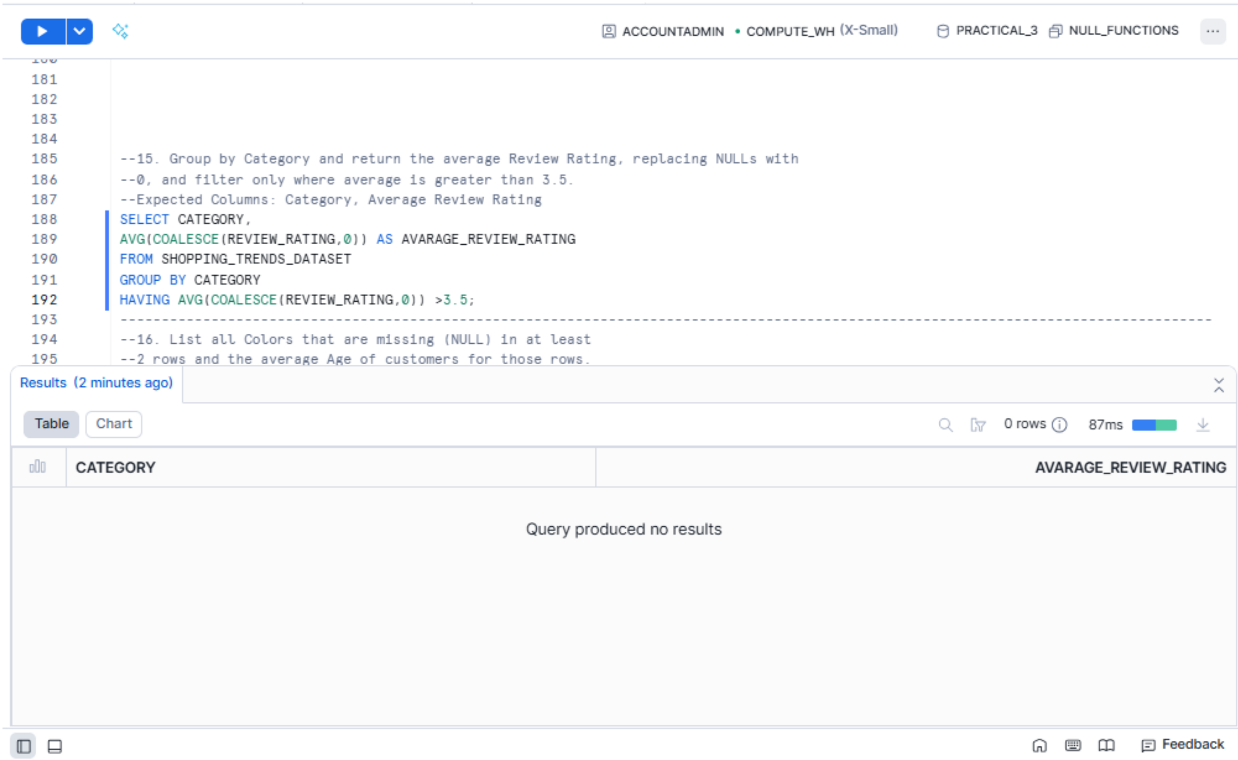
Results (just now)

Table Chart 7 rows 76ms

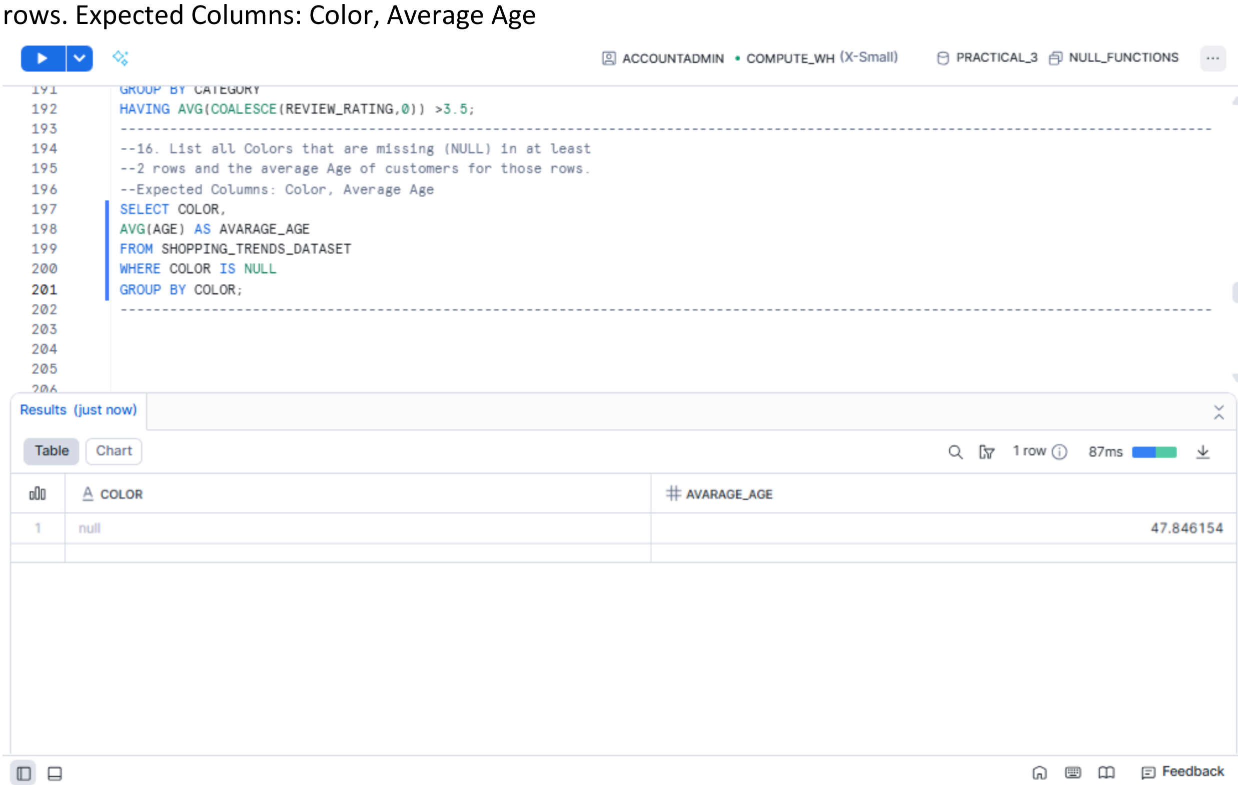
	PAYMENT_METHOD	MISSING_REVIEW_RATING_COUNT
1	Credit Card	0
2	Cash	0
3	null	0
4	Debit Card	0
5	Venmo	0
6	PayPal	0
7	Bank Transfer	0

Feedback

15. Group by Category and return the average Review Rating, replacing NULLs with 0, and filter only where average is greater than 3.5. Expected Columns: Category, Average Review Rating



16. List all Colors that are missing (NULL) in at least 2 rows and the average Age of customers for those rows. Expected Columns: Color, Average Age



17. Use CASE to create a column Delivery Speed: 'Fast' if Shipping Type is 'Express' or 'Next Day Air', 'Slow' if 'Standard', 'Other' for all else including NULL. Then count how many customers fall into each category.

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ACCOUNTADMIN • COMPUTE_WH (X-Small) | PRACTICAL_3 | NULL_FUNCTIONS

```sql
--18. Find customers whose purchase_amount is NULL and whose Promo Code Used is
--'Yes'.
--Expected Columns: Customer ID, purchase_amount, Promo Code Used -
SELECT CUSTOMER_ID,
PURCHASE_AMOUNT,
PROMO_CODE_USED
FROM SHOPPING_TRENDS_DATASET
WHERE PURCHASE_AMOUNT IS NULL AND PROMO_CODE_USED ='YES';
------------------------------------------------------------------------------------------------------------------------




--19. Group by Location and show the maximum Previous
```

Results (just now)

Table | Chart | 20 rows | 73ms

	CUSTOMER_ID	PURCHASE_AMOUNT	PROMO_CODE_USED
1	13	null	TRUE
2	30	null	TRUE
3	78	null	TRUE
4	95	null	TRUE
5	124	null	TRUE
6	129	null	TRUE
7	130	null	TRUE
8	138	null	TRUE

Feedback

18. Find customers whose purchase_amount is NULL and whose Promo Code Used is 'Yes'. Expected Columns: Customer ID, purchase_amount, Promo Code Used

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```sql
--19. Group by Location and show the maximum Previous
--Purchases, replacing NULLs with 0, only where the average rating is above 4.0.
--Expected Columns: Location, Max Previous Purchases, Average
--Review Rating
SELECT LOCATION,
MAX(COALESCE(PREVIOUS_PURCHASES,0))
AS MAXIMUM_PREVIOUS_PURCHASES,
AVG(REVIEW_RATING) AS AVERAGE_RATING
FROM SHOPPING_TRENDS_DATASET
GROUP BY LOCATION
HAVING AVERAGE_RATING > 4.0;
------------------------------------------------------------------------------------------------------------------------
```

Results (just now)

Table | Chart | 0 rows | 88ms

	LOCATION	MAXIMUM_PREVIOUS_PURCHASES	AVERAGE_RATING

Query produced no results

Feedback

19. Group by Location and show the maximum Previous Purchases, replacing NULLs with 0, only where the average rating is above 4.0. Expected Columns: Location, Max Previous Purchases, Average Review Rating

```sql
--19. Group by Location and show the maximum Previous
--Purchases, replacing NULLs with 0, only where the average rating is above 4.0.
--Expected Columns: Location, Max Previous Purchases, Average
--Review Rating
SELECT LOCATION,
MAX(COALESCE(PREVIOUS_PURCHASES,0))
AS MAXIMUM_PREVIOUS_PURCHASES,
AVG(REVIEW_RATING) AS AVERAGE_RATING
FROM SHOPPING_TRENDS_DATASET
GROUP BY LOCATION
HAVING AVERAGE_RATING > 4.0;
----------------------------------------------------------------------------------------------------------------------------
```

Results (26 minutes ago)

Table Chart 0 rows 88ms

LOCATION	MAXIMUM_PREVIOUS_PURCHASES	AVERAGE_RATING

Query produced no results

Feedback

20. Show customers who have a NULL Shipping Type but made a purchase in the range of 30 to 70 USD. Expected Columns: Customer ID, Shipping Type, purchase_amount, Item Purchased

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```sql
--20. Show customers who have a NULL Shipping
--Type but made a purchase in the range of 30 to 70 USD.
--Expected Columns: Customer ID, Shipping
--Type, purchase_amount, Item Purchased
SELECT CUSTOMER_ID,
SHIPPING_TYPE,
PURCHASE_AMOUNT,
ITEM_PURCHASED
FROM SHOPPING_TRENDS_DATASET
WHERE PURCHASE_AMOUNT BETWEEN 30 AND 70
AND SHIPPING_TYPE IS NULL;
```

Results (29 minutes ago)

Table Chart 0 rows 88ms

LOCATION	MAXIMUM_PREVIOUS_PURCHASES	AVERAGE_RATING

Query produced no results

Feedback